

Features that stand out

- Virtual lab to design and carry out simulations of electronic circuits
- Includes tutorials and learning guides to start the use of the program
- Assorted examples of templates that you can adapt to your projects
- Contains all kinds of components to carry out the tests: coils, resistors, transistors and integrated circuits

Courtesy: livewire.en.malavida.com

even while the circuit is simulating, like, may be, increasing resistance of a potentiometer.

One might argue that this is the norm, but circuits that work during simulation might not work well in hardware. Livewire tries to cast this doubt aside by incorporating an advanced physics engine into the simulation routine; the result is fast and accurate, and can be relied upon.

A range of resources to choose from. It is easy to put your plan into action owing to the availability of a vast library of simulated components, ranging from simple resistors to sophisticated integrated circuits. There are discrete semiconductors like diodes or transistors, connectors, power supplies, passive components and ones that can be used as inputs and outputs.

If you are in need of more, just drop the team a mail and they would consider including your suggestions, claims the website.

Real-world models help you put together a circuit that would be replicated in the hardware later. For theoretical models that are still in the idea stage, each component can be replaced with its virtual model, helping you explore and groom the innovative thought. You can even simulate faults in components or component destruction—be at your

pessimistic best!

Make every second count. To aid the process of simulation, there is also a collection of seven virtual instruments that mimic real-time functions and measurements. The list includes an oscilloscope to record voltage over time, a multimeter to measure current, voltage and resistance, a logic analyser to produce graphs of logic levels, a wattmeter to measure power reading, a signal generator to produce waveforms and a word generator to drive current by generating streams of 16-bit words.

The circuit at its stylish best

In addition to the above, you can even choose to view your design in different styles from the preset list,

allowing you to alter the way your circuit looks while simulating. The styles, in reality, are a boon and even make the design process very interesting. If you choose voltage-level view, a bar opens up next to each component, its height proportional to the instantaneous voltage value across the component. The charge on a capacitor keeps moving animatedly; more the plus or minus signs on the capacitor plates, more is the stored charge.

Monitoring current spikes. An arrow on a wire shows the direction of current through it, and if you move the cursor to it, instantaneous current reading pops up. Current-flow view truly feels like a mini video game; small dashes moving along wires indicating movement of current, maintaining alternating current direction changes and switch patterns. Looking more like a dance routine, it is very pleasing to the eye.

Between highs and lows. On choosing logical-level view, you can actually view voltages, current, charge and even digital logic levels.

Logical high and low are shown in different colours, looking like a constantly blinking pattern with voltage level indicated next to it.

Normal view certainly looks very normal in front of the other interesting views. A nice way to get into serious designing, would you not say?

Make your PCB on the go

If your end idea is to put your design together on a printed circuit board (PCB), your job is made that much easier with Livewire. This software is tightly integrated with PCB Wizard 3, a PCB

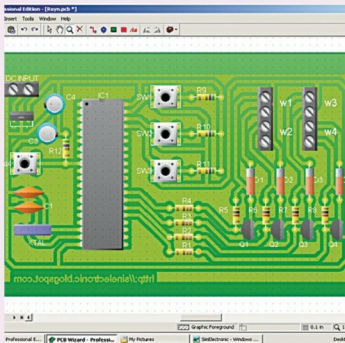


Fig. 2: Translating a design to a PCB (Image courtesy: www.fortex.co.uk/product/livewire)